typst:

```
$
  &not (p and (q or r)) \
  &= not p or (not q and not r) & "(demorgan twice)" \
  &= (not p or not q) and (not p or not r) wide &
"(distributive)" \
  &= not (p and q) and not(p and r) & "(demorgan)"
$
```

latex:

```
\begin{aligned}
  & \neg (p \land (q \lor r)) \\
  &= \neg p \lor (\neg q \land \neg r) & \text{(demorgan
twice)} \\
  &= (\neg p \lor \neg q) \land (\neg p \lor \neg r)
\qquad & \text{(distributive)} \\
  &= \neg (p \land q) \land \neg(p \land r) &
\text{(demorgan)}
\end{aligned}
```

$$\begin{aligned}
& \neg (p \land (q \lor r)) \\
&= \neg p \lor (\neg q \land \neg r) & \text{(demorgan twice)} \\
&= (\neg p \lor \neg q) \land (\neg p \lor \neg r) \qquad & \text{(distributive)} \\
&= \neg (p \land q) \land \neg(p \land r) & \text{(demorgan)}
\end{aligned}$$